Master's Thesis Proposal

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Contents

1 Introduction and Background of the Study

1.1 The Evolution of Video Understanding: From Perception to Reasoning

The field of Computer Vision has witnessed a metamorphic transformation over the past decade. Historically, the discipline was concerned with Perceptionthe ability of a machine to recognize distinct visual patterns, such as edges, textures, and specific object classes (e.g., "car," "pedestrian"). Early methodologies relied on handcrafted features and heuristic algorithms, which were severely limited by their inability to generalize to unseen environments. The advent of Deep Convolutional Neural Networks [Krizhevsky et al., 2017, CNNs] in the early 2010s marked the first major paradigm shift, enabling robust object detection and semantic segmentation based on visual taxonomy.

However, the current decade has ushered in a second, more profound paradigm shift: the transition from Perception to Reasoning. In complex, real-world environments, identifying an object by its class is often insufficient. For instance, in an autonomous driving scenario, a system must distinguish between "the pedestrian waiting on the curb" and "the pedestrian about to cross against the light." Visually, they may look identical; the distinction lies in the temporal context and implicit intent. This necessity has gave rise to the field of Reasoning Video Segmentation (RVS).

Unlike traditional Video Object Segmentation (VOS), which typically relies on an initial mask provided by a human user (semi-supervised) or simple category labels (unsupervised), RVS integrates Multimodal Large Language Models (MLLMs) into the visual pipeline. Systems such as [Lai et al., 2024, LISA] (Reasoning Segmentation via Large Language Model) and its video successors, [Bai et al., 2024, VideoLISA] and [Zheng et al., 2025, ViLLa], allow users to query video data using complex, natural language expressions that require logical deduction, spatial awareness, and causal reasoning. For example, a query might be: "Segment the vehicle that caused the traffic jam." To execute this, the model must not only recognize vehicles but also understand the concept of a "traffic jam," identify the causal agent, and track that specific agent across time.

1.2 The Technological Backbone: Multimodal LLMs and Vision Transformers

The current state-of-the-art in RVS is built upon massive Foundation Models. Typically, these architectures follow a tripartite structure:

1. The Visual Encoder: Utilizing architectures like [Radford et al., 2021, CLIP] or the [Ravi et al., 2024, SAM 2] (Segment Anything Model 2) image encoder to compress high-resolution video frames into abstract visual embeddings.

2. The Reasoning Core: A Large Language Model (e.g., [Li et al., 2024, LLaVA] or [Zheng et al., 2023, Vicuna]) that processes the user's textual query and attends to the visual embeddings to "reason" about which object matches the description.

3. The Mask Decoder: A specialized module that translates the LLMs decision into a binary pixel mask.

As of late 2025, [Ravi et al., 2024, SAM 2] represents the pinnacle of this technology. It introduced a "streaming memory" architecture allowing the model to store information about an object in a memory bank, theoretically enabling it to handle occlusions and re-appearances. These advancements have pushed the semantic accuracy of video segmentation to unprecedented levels, allowing machines to understand nuanced human instructions.

1.3 The Theoretical Conflict: Stochastic Deep Learning vs. Deterministic Kinematics

Despite these advances, a fundamental theoretical conflict remains at the heart of modern video segmentation. Deep Learning models are inherently stochastic and data-driven. They treat every video frame as a new statistical puzzle to be solved. While attention mechanisms provide some temporal context, the model effectively "re-imagines" the object in every frame based on learned feature correlations. In contrast, the physical world is deterministic and kinematic. Objects in motion follow the laws of physicsmomentum, inertia, and continuous trajectories. A car cannot teleport five meters to the left between Frame t and Frame $t+1$ However, purely Deep Learning-based models often produce outputs that violate these physical laws, resulting in "temporal jitter" (flickering masks) or "hallucinations" (masks appearing in empty space).

This thesis posits that the reliance on purely Deep Learning approaches for tracking is fundamentally inefficient. While Deep Learning is essential for the semantic reasoning (knowing "what" to track), Classical Mechanics and Geostatistics are far better suited for the spatial tracking (knowing "where" it will go). This study explores the intersection of these two fields, proposing that Spatio-Temporal Interpolation (specifically Kriging and Radial Basis Functions) and Trajectory Forecasting ([Hochreiter and Schmidhuber, 1997, LSTM]/ [Oreshkin et al., 2020, N-BEATS]) can stabilize the stochastic outputs of modern AI.

1.4 The Resource Crisis in Edge Computing

The background of this study is also defined by a hardware crisis. The computational cost of running models like [Bai et al., 2024, VideoLISA] or [Ravi et al., 2024, SAM 2] s immense. A standard inference pass requires calculating attention matrices that scale quadratically with sequence length ($O(N^2)$). For a drone or an autonomous robot operating on battery power with limited thermal headroom, running a billion-parameter model at 30 Frames Per Second (FPS) is physically impossible.

2 Statement of the Problem

The rapid advancement of Reasoning Video Segmentation (RVS) has created a dichotomy where semantic capability has outpaced temporal stability and computational efficiency. While we can now ask machines to perform complex visual reasoning tasks, the implementation of these systems in real-world, real-time scenarios is obstructed by three critical, interrelated problems.

2.1 Problem 1: The Computational Inefficiency of Dense Sampling

The first and most pragmatic problem is the prohibitive computational cost of Dense Sampling Strategies. Current state-of-the-art models, including [Bai et al., 2024, VideoLISA] and [Zheng et al., 2025, ViLLa], are designed to process video data frame-by-frame. In this paradigm, the Multimodal Large Language Model (MLLM) and the Vision Encoder must perform a full forward pass for every single image in the video sequence.

Given that a standard video operates at 30 to 60 FPS, and a single inference of an MLLM can take between 50ms to 200ms on high-end consumer hardware (e.g., NVIDIA RTX 4090), achieving real-time performance is mathematically impossible without massive hardware scaling. For edge applicationssuch as Search and Rescue Drones, Agricultural Robotics, or wearable AR devicesthis latency is unacceptable. These devices typically operate on embedded chips (e.g., NVIDIA Jetson Orin) with a fraction of the compute power of a data center GPU. Consequently, the current RVS framework is confined to offline processing, rendering it useless for applications requiring immediate decision-making.

2.2 Problem 2: Temporal Jitter and Semantic Flickering

The second problem is the phenomenon of Temporal Jitter, also known as mask flickering. Deep Learning segmentation models typically process frames with a high degree of independence. Even with mechanisms like "Memory Attention" in [Ravi et al., 2024, SAM 2], the model re-evaluates the pixel-wise classification probabilities at each timestep.

Because neural networks are sensitive to minor perturbations in input data (pixel noise, lighting changes, slight motion blur), the generated mask often exhibits high-frequency fluctuations in its boundary shape. An object that is rigid in the real world (e.g., a truck) might appear to "wobble" or deform in the segmented video. This is not merely an aesthetic issue; in downstream applications like robotic grasping or collision avoidance, these fluctuations introduce noise into the control loop, potentially leading to system failure. The problem stems from the fact that standard RVS models maximize Semantic Accuracy (IoU) on single frames but do not explicitly optimize for Kinematic Smoothness across time.

2.3 Problem 3: Tracking Failure During Occlusion (The "Object Permanence" Deficit)

The third and most complex problem is the failure of attention mechanisms during Occlusions. In video data, objects are frequently obscured by foreground elements (e.g., a pedestrian walking behind a tree).

Modern Transformers rely on "Attention" to track objects. Attention calculates the similarity between the object's features in the current frame and its features in memory. However, when an object is fully occluded, its visual features disappear. The Attention mechanism effectively "loses sight" of the target. While models like [Ravi et al., 2024, SAM 2] attempt to handle this with a memory bank, they often struggle with Long-Term Occlusion. If the object remains hidden for 20 or 30 frames, the attention weights decay, and the model "forgets" the object exists. When the object re-emerges, the model often treats it as a new, unrelated entity (ID Switch).

This failure occurs because Transformers lack an explicit Motion Model. They do not intuitively understand "Object Permanence" or momentum. Unlike a human, who can predict where a car will emerge from a tunnel based on its speed before entering, a Transformer only knows what it can "see." This lack of predictive capability leads to segmented tracks that are broken, discontinuous, and unreliable in dynamic environments.

3 Research Aim and Hypothesis

3.1 Research Aim

The overarching aim of this research is to develop, implement, and validate a Hybrid Sparse-Sampling Architecture for Reasoning Video Segmentation. This architecture seeks to decouple the heavy semantic processing from the spatial tracking task.

Specifically, the aim is to create a system that utilizes Deep Learning ([Bai et al., 2024, VideoLISA]/[Ravi et al., 2024, SAM 2]) only on sparse keyframes to establish semantic understanding, while employing lightweight Classical Interpolation (Kriging) and Trajectory Forecasting ([Hochreiter and Schmidhuber, 1997, LSTM]) to reconstruct the intermediate frames. By doing so, the research aims to prove that it is possible to achieve State-of-the-Art (SOTA) accuracy and stability on resource-constrained hardware, effectively solving the "Stability-Efficiency Trade-off."

3.2 Research Hypothesis

The study is guided by the following core hypothesis and sub-hypotheses:

Core Hypothesis:

It is hypothesized that a hybrid segmentation framework, which combines sparse semantic keyframe extraction with geostatistical interpolation and LSTM-based trajectory forecasting, will achieve higher temporal stability and computational efficiency than a purely deep-learning-based dense sampling approach, without a statistically significant reduction in semantic accuracy.

Sub-Hypothesis 1 (Efficiency):

By reducing the frequency of deep learning inference to 20% of the video frames (Sparse Sampling) and utilizing interpolation for the remaining 80%, the system will achieve a processing speed increase of at least 200% (3x speedup) compared to the baseline [Bai et al., 2024, VideoLISA] model.

Sub-Hypothesis 2 (Stability):

The application of Geostatistical Interpolation (Universal Kriging) on mask contours will result in a lower "Flow-Warping Error" (a metric of temporal jitter) compared to masks generated independently by neural networks, due to the variance-minimizing properties of the Kriging estimator.

Sub-Hypothesis 3 (Occlusion):

An LSTM-based motion predictor, trained on object centroid trajectories, will maintain higher Intersection-over-Union (IoU) scores during periods of full occlusion (5+ frames) compared to the standard visual attention mechanisms of Transformers, as the LSTM explicitly models momentum and velocity.

4 Research Objectives

To achieve the stated aim and test the hypotheses, this research is structured around four distinct, measurable objectives.

4.1 Objective 1: To Design a Sparse-Sampling Framework for Reasoning Segmentation

The first objective is to architect the software pipeline that allows for the decoupling of "Reasoning" and "Tracking."

- **Rationale:** Current models are monolithic; they cannot easily skip frames without losing context. I must design a modular system where the "Semantic Module" ([Bai et al., 2024, VideoLISA]) and the "Interpolation Module" are distinct but synchronized.
- **Implementation:** This involves developing a Keyframe Selector Algorithm. Rather than selecting keyframes arbitrarily (e.g., every 5th frame), the algorithm will analyze pixel difference metrics (Optical Flow magnitude) to trigger the heavy Deep Learning model only when the scene changes significantly. This ensures that compute resources are invested intelligently.

4.2 Objective 2: To Develop a Lightweight Trajectory Forecasting Module

The second objective is to instill the system with a sense of "physics" and "object permanence."

- **Rationale:** To survive occlusions and reduce sampling rates, the system must be able to "hallucinate" the future position of an object.
- **Implementation:** I will design and train a Recurrent Neural Network (RNN), specifically utilizing Long Short-Term Memory ([Hochreiter and Schmidhuber, 1997, LSTM]) or N-BEATS units. This model will take a lightweight input vector (Centroid x, y Bounding Box w, h) rather than heavy image features. It will be trained on the YouTube-VOS trajectory dataset to learn typical motion patterns (linear, arc, acceleration). This module will act as the "autopilot" for the system between keyframes.

4.3 Objective 3: To Adapt Geostatistical Kriging for Efficient Mask Reconstruction

The third objective addresses the mathematical challenge of generating the mask shape without running a neural network.

- **Rationale:** While the [Hochreiter and Schmidhuber, 1997, LSTM] predicts where the object is, it does not describe shape. Linear interpolation is too rigid for organic objects (e.g., a running dog).

- **Implementation:** I will implement Universal Kriging and Radial Basis Function (RBF) interpolation. Crucially, to avoid the $O(N^3)$ computational complexity of Kriging, this objective includes the development of a "Contour-Vectorization" method. We will convert the binary raster masks from the keyframes into sparse polygon vertices, apply Kriging only to these vertices to model their deformation over time, and then rasterize the result. This transforms a pixel-heavy problem into a lightweight vector problem.

4.4 Objective 4: To Validate the System via Comprehensive Benchmarking

The final objective is to provide empirical evidence of the system's performance relative to the current State-of-the-Art ([Ravi et al., 2024, SAM 2] 2 / [Bai et al., 2024, VideoLISA]).

- **Rationale:** Theoretical novelty is insufficient; the system must work on standard datasets to be academically valid.
- **Implementation:** The system will be tested on the MeViS (Motion Expressions Video Segmentation) dataset for reasoning capability and the DAVIS dataset for occlusion handling. I will measure:
  1. **Semantic Accuracy:** Using the Jaccard Index (J) and F-score (F)
  2. **Temporal Stability:** Using the Temporal Consistency Score ($T_s tab$)
  3. **Efficiency:** Measuring FPS and Peak VRAM usage on a consumer-grade GPU (simulating edge constraints).

5 Main References

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